



NotifiED

Mobile AI-Assisted Personal Campus Information Hub

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QUALITY
EDUCATION



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INDUSTRY, INNOVATION
AND INFRASTRUCTURE



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Introduction

In many schools, effective communication between students, faculty, and organizations is essential for keeping everyone informed and engaged in campus activities. While announcements are widely shared through platforms such as social media, messaging apps, and email, each organization typically uses its own channel. Because of this, students often struggle to identify where to find official updates or which platform to follow for specific organizations.

As a result, important information, events, and opportunities are overlooked not due to lack of availability, but because of the absence of a unified access point. At the same time, faculty members and student organizations face challenges in ensuring that their announcements consistently reach their intended audience.

To address this issue, this project introduces NotifiED, an AI-assisted mobile inquiry module designed to help students manage fragmented campus communication channels within educational institutions. Rather than replacing existing communication systems, the application functions as a personalized digital secretary that retrieves and organizes campus-related information from a student's authorized platforms such as institutional email accounts, messaging applications, and organization social media pages. The system provides students with a unified and accessible interface where they can view announcements, explore organizations, browse campus events, and receive AI-assisted summaries of important updates without manually searching across multiple platforms. In addition, NotifiED features an AI-powered chatbot that assists users by answering questions, summarizing announcements, and guiding them through relevant campus information. The application also incorporates automated push notifications and scheduling support features that help students stay informed about activities that match their availability, ultimately making campus engagement more organized, efficient, and personalized.

Project Context

NotifiED: Mobile AI-Assisted Personal Campus Information Hub is a scalable, mobile-based inquiry application designed to improve how students access and interact with campus-related information. Rather than replacing existing institutional communication systems, the application focuses on retrieving, organizing, and presenting campus information from multiple communication channels into a unified and user-friendly mobile interface. Through AI-assisted retrieval and summarization features, the system helps students conveniently access announcements, explore events, and manage relevant campus information without manually monitoring multiple platforms.

The application is initially implemented within Asia Pacific College (APC) as a testing ground to evaluate its usability, accessibility, and overall effectiveness in addressing communication challenges among students. The long-term objective is to develop a scalable solution that can be adapted and deployed across multiple educational institutions.

Target Users:

The system is designed to serve the following primary users:

- Students (Primary Users):
Students serve as the main users of the application. They are provided with a personalized mobile interface where they can conveniently access announcements, discover campus events, and manage relevant schedule information retrieved from authorized communication platforms. The system is

designed to minimize the need to manually search across multiple platforms, thereby improving accessibility, organization, and engagement.

- **Campus Organizations and Administrators (Information Sources):** These groups act as sources of information by providing announcements, event details, and updates that are made accessible through the platform. Their contributions ensure that students receive relevant and timely information.

The Current System

At present, students encounter difficulties in engaging with campus activities due to the decentralized nature of information dissemination. Announcements and event details are distributed across multiple platforms, including social media, messaging applications, email, and physical bulletin boards.

While this approach increases the visibility and reach of information, it also introduces several challenges:

- **Platform Ambiguity:** Each organization utilizes its own preferred communication channel, making it difficult for students to determine where to access official updates or which platforms to follow.
- **Inconsistent Access to Information:** Despite the wide availability of announcements, the absence of a unified access point makes it challenging for students to stay consistently informed.
- **Limited Accessibility:** There is no single interface where students can conveniently view all relevant and verified information.

As a result, students may miss important announcements and opportunities not due to a lack of information, but because of the absence of a centralized and accessible system that consolidates these updates effectively.

Project Goal

The primary goal of this project is to develop an AI-assisted, student-centered mobile inquiry application that enhances access to campus-related information and improves overall user experience. The system aims to help students efficiently retrieve, organize, and interact with announcements, events, and organizational updates gathered from multiple existing communication platforms through a unified and personalized mobile interface.

Alignment with Institutional Goals

Educational institutions continuously aim to enhance communication efficiency, increase student engagement, and adopt digital solutions to improve operational processes. NotifiED supports these objectives by:

- Providing students with a unified mobile interface for accessing campus-related information retrieved from multiple communication platforms
- Improving student awareness by simplifying the retrieval and organization of announcements and event updates from existing institutional channels
- Minimizing communication gaps caused by fragmented and platform-dependent information dissemination
- Supporting the integration of AI-assisted and user-centered technologies within educational environments without requiring institutions to replace their current communication systems

Contribution to Long-Term Vision

This project contributes to the long-term vision of establishing a more connected and technologically adaptive educational environment. By providing students with an AI-assisted interface that organizes campus-related information from existing communication channels, NotifiED improves students' ability to stay informed and participate in institutional activities more efficiently.

Furthermore, the system is designed with scalability in mind, allowing the inquiry framework to be adapted and implemented across various educational institutions. As a mobile-based AI-assisted retrieval solution, the project demonstrates potential for broader adoption in improving how students manage fragmented campus communication systems while maintaining compatibility with existing institutional platforms.

Statement of the Problem

Effective communication is essential within educational institutions to ensure that students remain informed and actively engaged in campus activities. However, many schools rely on multiple independent communication platforms for disseminating institutional information. Different student organizations and departments often use their own preferred channels, such as social media platforms, email services, messaging applications, and group chats, to distribute announcements and event updates. As a result, students frequently have trouble identifying where to access official information or which platforms to monitor for specific organizations and campus-related updates. This fragmented communication environment creates inconsistency in information accessibility and makes it challenging for students to remain consistently informed about campus activities.

In response to these issues, this study proposes the development of an AI-assisted mobile inquiry application that helps students retrieve, organize, and access campus-related information from multiple existing communication channels through a unified and personalized mobile interface.

Specifically, the study seeks to address the following problems:

Problem 1: Difficulty Managing Fragmented Campus Information

There is no intelligent system that helps students efficiently organize and access campus-related announcements, events, and updates distributed across multiple communication platforms. As a result, different departments and student organizations continue to rely on their own preferred channels, such as social media, emails, group chats, and bulletin boards, for information dissemination.

This fragmented setup creates multiple challenges for students, including difficulty identifying where to access official information, the need to manually monitor multiple platforms, and exposure to excessive and uncoordinated content. Consequently, communication becomes inconsistent, important announcements are overlooked, and some events receive limited visibility. These factors contribute to reduced student awareness, lower engagement, and missed opportunities for participation in campus activities.

Problem 2: Difficulty in Accessing Event Information

Students often encounter challenges in efficiently accessing complete, organized, and relevant details regarding campus events. Because event-related

information is distributed across multiple independent communication platforms, students need to manually search through emails, social media posts, messaging groups, and organization pages to gather important details. This fragmented retrieval process can result in inconvenience, delayed access to information, and reduced participation in campus activities.

Problem 3: Poor User Experience in Existing Information Channels

Current platforms used for campus communication, such as social media applications, emails, and messaging services, are not specifically designed to organize and personalize campus-related information for students. As a result, students often have trouble navigating through large volumes of unrelated or unstructured content, identifying relevant updates, and efficiently retrieving important announcements.

This lack of intelligent organization and user-centered information presentation leads to frustration, reduced efficiency in information retrieval, notification fatigue, and decreased engagement with campus-related content.

Objectives

Main Objective

The primary objective of this study is to develop an AI-assisted mobile inquiry application that improves how students access and manage campus-related information by retrieving, organizing, and presenting announcements, events, and institutional updates from multiple existing communication platforms through a unified and personalized mobile interface.

Specific Objectives

1. To develop an AI-assisted mobile inquiry platform for campus information access
To design and implement a mobile application that retrieves and organizes school-related announcements, events, and updates from existing communication channels, reducing platform ambiguity and improving accessibility of institutional information for students.
2. To improve the accessibility and clarity of event information
To present complete, organized, and easily understandable event details within the application through AI-assisted retrieval and information organization, minimizing confusion caused by fragmented communication sources and reducing the need for students to manually monitor multiple platforms.
3. To enhance user experience in accessing campus information
To develop a user-friendly and personalized mobile interface that allows students to efficiently navigate retrieved campus information, access relevant updates, and stay informed with minimal effort, thereby improving engagement with campus activities and reducing information overload.

Significance of the Project

The primary beneficiaries of the NotifiED project are the students who serve as the main users of the application. The system provides students with a personalized and organized mobile interface where campus-related announcements, events, and updates retrieved from existing communication platforms can be conveniently accessed without the need to manually monitor multiple channels. This helps reduce information overload, improve accessibility to relevant institutional information, and create a more organized campus information experience.

If this project is successful, it may help address the challenges caused by fragmented campus communication by providing students with a more efficient way of

retrieving and organizing information from multiple platforms. Instead of requiring students to constantly switch between different communication applications, the system enhances accessibility through AI-assisted retrieval, intelligent organization, and personalized notification features, which may improve awareness and participation in institutional activities. Additionally, NotifiED supports United Nations Sustainable Development Goal 4: Quality Education by improving equitable access to educational and institutional information, while also supporting Sustainable Development Goal 9: Industry, Innovation, and Infrastructure through the integration of AI-assisted and mobile-based technologies that enhance digital communication and information accessibility within educational environments.

Scope and Limitations

This section outlines the specific focus and boundaries of the NotifiED project. The project is designed to develop a functional AI-assisted mobile inquiry application that helps students retrieve, organize, and access campus-related information from multiple communication platforms while maintaining a focused set of features to ensure system efficiency.

Scope

Specific Field of Study: This project is situated within the fields of Information Technology and Mobile Application Development. It addresses communication challenges in educational institutions by developing an AI-assisted mobile inquiry system that organizes and presents campus-related information retrieved from existing communication channels through a personalized student interface.

Fundamental Concepts: The core principles of this study are AI-assisted information retrieval, information organization, and real-time information accessibility. The

system aims to reduce the inefficiencies caused by fragmented communication channels by helping students efficiently retrieve and access announcements, event-related updates, and campus information from multiple existing platforms.

Target Users and Beneficiaries: The primary users and beneficiaries of the application are students who utilize the system to access announcements, browse events, explore organizations, and receive campus-related updates retrieved from authorized communication platforms. The system focuses specifically on improving the student-side experience of accessing and organizing institutional information.

Target Environment: The platform is designed for educational institutions, specifically targeting academic environments such as colleges and universities. It provides a structured and dedicated mobile interface that assists students in managing campus-related information gathered from existing communication systems.

System Output: The output of this study is a functional mobile inquiry application that retrieves, organizes, summarizes, and presents campus-related announcements, events, and organization updates through an AI-assisted and student-centered mobile interface.

Limitations

To ensure the successful completion and reliability of the system, the following boundaries have been established:

Institutional Scope: The system is strictly limited to educational institutions. It is not intended for commercial use, e-commerce, or general-purpose social networking.

Connectivity Requirements: The application requires a stable internet connection to retrieve and process campus-related information from authorized communication platforms. Offline functionality is not supported.

Supplementary Nature: The system serves as a supplementary inquiry and information organization tool. It does not replace official institutional communication systems, academic portals, grading systems, or formal institutional emails.

Development Focus: This project focuses specifically on the frontend mobile inquiry experience and AI-assisted information retrieval features for students. The development and maintenance of institutional backend infrastructures, administrative systems, servers, and databases are outside the implementation scope of this study.

Functional Boundaries: The platform is designed solely for retrieving, organizing, and presenting campus-related information for students. It does not include private peer-to-peer messaging, financial transaction processing, institutional record management, or direct administrative posting management within the application.

Privacy and Access Limitations: The system only accesses communication platforms and information streams explicitly authorized by the user. Retrieved information is processed solely for personalized organization and viewing within the user's application environment.

Review of Related Literature / Systems

The development of NotifiED exists within a broader academic and technological context that underscores both the urgency and viability of centralized, AI-assisted campus communication platforms. This chapter reviews existing literature and studies relevant to the core themes of the project, namely: the challenges posed by fragmented digital communication in educational settings, the role of mobile technology in improving student engagement and institutional operations, the application of artificial intelligence in educational platforms, and the fatigue introduced by disorganized multi-platform environments. Together, these bodies of literature establish the conceptual and empirical foundations upon which NotifiED is built.

The Problem of Fragmented Campus Communication A recurring finding in recent literature is that decentralized information dissemination creates systemic communication barriers in educational institutions. Cheong and Nyaupane (2022) examined how smart campus applications have been adopted across universities to consolidate student services into connected digital portal experiences, noting that networked systems are recognized for their role in sustaining student security, well-being, and community engagement. Their work surfaces the growing institutional demand for unified digital infrastructure, a demand directly mirrored in the communication gaps that motivated NotifiED's design. Complementing this, Orhani et al. (2024) found that integrated ICT platforms reduce administrative friction by automating routine tasks such as event announcements and communications, ensuring that information reaches the appropriate audience in a timely and organized manner. This consolidation, they argue, is no longer a matter of preference but a prerequisite for operational quality in modern educational institutions. Hidayat and Farhan (2025) further establish that the fragmentation of announcements across informal channels such as Messenger, Viber, and Facebook generates significant cognitive load among students, contributing to information avoidance and increased academic anxiety, a phenomenon they describe as digital clutter that buries critical updates beneath unrelated social content.

Digital Fatigue as a Barrier to Student Engagement Parallel to the fragmentation problem, literature from the Philippine context highlights the psychological toll of navigating multiple, uncoordinated digital platforms. Tolentino and Miranda (2024) documented significant levels of general and visual fatigue among Filipino students in state universities caused by the constant need to switch between different interfaces and manage multiple digital feeds, a condition they associate with reduced motivation and diminished academic involvement. Oducado et al. (2024) similarly found a high prevalence of digital learning fatigue among Filipino nursing students, demonstrating a significant correlation between platform overload and lower online learning satisfaction, a condition that worsens as the number of uncoordinated digital streams a student must manage increases. These Philippine-specific findings lend local grounding to the global problem of digital burnout and reinforce the argument that consolidating campus information into a single, non-redundant platform is not merely a matter of convenience but a measurable contributor to student well-being and academic performance.

Mobile Applications and Student Engagement in Higher Education On the technological response side, studies consistently affirm the capacity of well-designed mobile applications to improve student engagement and institutional responsiveness. Ibrahim (2024) examined mobile applications across ICT and computer science education contexts, identifying positive correlations between specific application functionalities and heightened student engagement levels, while cautioning that benefits are contingent on thoughtful, user-centered design rather than technology deployment alone. A 2023 national survey by Inside Higher Ed and College Pulse further validated this direction, reporting that students as digital natives actively seek campus technology that fosters connection and reduces the friction of accessing institutional information. Hossain et al. (2023) demonstrated at the departmental level that Android-based university applications improve coordination, promote transparency, and provide round-the-clock accessibility for both students and administrators, effectively replacing fragmented manual communication systems with a more responsive and organized digital alternative.

AI-Assisted Communication Tools in Education The integration of artificial intelligence further distinguishes NotifiED from conventional centralized platforms. Wollny et al. (2021), in a systematic review of 74 publications on chatbots in education, identified that AI-powered chatbots offer scalable and accessible possibilities as communication and information tools for digital learning environments, serving objectives that include skill support, education efficiency through task automation, and the provision of time-saving services for students. The vmedulife Software (2025) discussion on centralized communication tools reinforces this architecture from a governance standpoint, arguing that a centralized information hub functions as a single source of truth for all campus constituents, minimizing conflicting information and enabling a structured, verified flow of announcements and organizational data.

Synthesis and Conclusion Collectively, the literature reviewed in this chapter presents a converging body of evidence in support of NotifiED's core design rationale. The fragmentation of campus communication across multiple platforms has been empirically linked to cognitive overload, digital fatigue, missed announcements, and reduced student engagement, problems documented both globally and specifically within the Philippine higher education context. The literature equally demonstrates that centralized mobile platforms, when designed with user-centered principles, measurably improve engagement and institutional efficiency, while AI-assisted features extend those platforms' responsiveness beyond static information delivery. These findings directly inform NotifiED's architecture: its centralized announcement feed addresses documented fragmentation and cognitive load issues; its mobile-first interface aligns with established evidence on student technology engagement; and its AI-powered chatbot responds to the literature's consistent identification of intelligent, scalable communication tools as a meaningful enhancement to student support. The combined evidence positions NotifiED not as a novel experiment but as a research-grounded response to a well-documented and locally validated institutional problem.

Current System

Technical Background

The current communication landscape within the institution is decentralized and relies on a mix of third-party platforms and traditional physical methods. This environment relies on multiple independent communication channels, leading to the fragmented information experience currently observed among students.

1. Hardware and Infrastructure:

- **Personal Computing Devices:** The primary hardware used by students and staff to access school-related updates through personal smartphones, tablets, and laptops.
- **Physical Information Boards:** Traditional corkboards and walls located in high-traffic areas (e.g., cafeteria, hallways) used for displaying printed posters and flyers.
- **Institutional Network Infrastructure:** The existing campus Wi-Fi that facilitates user access to external social media and messaging applications while on school grounds.

2. Software and Digital Platforms

- The current setup utilizes several "siloeed" digital platforms that operate independently and are not integrated into a unified student-centered information experience:
- **Social Media (Facebook/Instagram):** Student organizations and departments use dedicated pages for broadcasting. Official updates are often lost in "overflowing feeds" due to platform algorithms.

- Collaboration and Messaging (Microsoft Teams/Messenger): Important logistics and event details are frequently buried in various group chats, making it difficult for users to retrieve specific data later.
- Email Services (Microsoft Outlook): Administrators and organizations send mass email blasts, which contribute to "notification fatigue" and the risk of critical updates being overlooked.
- Cloud Storage (SharePoint/Google Drive): Used for internal document storage by organizations, but these files are rarely accessible to the general student body for information discovery.

3. Peopleware (Human Actors)

- Information Sources (Admins/Organizations): Responsible for manually creating promotional materials and cross-posting them across multiple uncoordinated platforms.
- Primary Users (Student Body): Acts as the end-user who must manually search for information across different applications and physical boards to stay informed.
- Oversight (School Administration): Provides general institutional oversight while relying on multiple communication platforms for information dissemination, which may contribute to inconsistencies in how students access campus-related information.

4. Network and Communication Methods

- Platform Ambiguity in Information Sources: There is no clearly defined reference point for official announcements, as each organization utilizes its own communication platform. As a result, students often do not know where to access specific updates, leading to difficulty in consistently staying informed.
- Manual Information Retrieval: Students must rely on word-of-mouth or their own individual effort to piece together a clear understanding of the campus schedule and active events.

List of Processes

Table 1 List of Processes in the Current System

Process ID	Process Name	Process Details
P001	Check Social Media Platforms	1. Student opens Facebook or Instagram. 2. Student scroll through organization pages and personal feeds. 3. Student searches for announcements or event posts. 4. Student attempts to identify which organization posted relevant updates.
P002	Check Messaging Applications	1. Student opens Messenger or similar messaging apps. 2. Student scroll through multiple group chats. 3. Student searches for event details or announcements buried in chat history. 4. Student attempts to piece together complete event information from fragmented messages.
P003	Check Email and Teams	1. Student opens Microsoft Outlook or Microsoft Teams. 2. Student sorts through mass email blasts and group notifications. 3. Student searches for specific announcements among unrelated messages. 4. Student flags or saves relevant updates to revisit later.
P004	Check Physical Bulletin Boards	1. Student visits high-traffic areas such as hallways or the cafeteria. 2. Student reads TV monitors. 3. Student notes event details manually or photographs the poster. 4. Student determines whether posted information is still current or outdated.

P005	Manual Information Retrieval and Verification	1. Student asks classmates or organization members for updates through word-of-mouth. 2. Student cross-references information gathered from multiple platforms. 3. Student determines whether the collected information is complete and accurate. 4. Student decides whether to attend or participate based on available information.
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Check Social Media Platforms

1. Student opens Facebook or Instagram.
2. Student scrolls through organization pages and personal feeds.
3. Student searches for announcements or event posts.
4. Student attempts to identify which organization posted relevant updates.

Check Messaging Applications

1. Student opens Messenger or similar messaging apps.
2. Student scroll through multiple group chats.
3. Student searches for event details or announcements buried in chat history.
4. Student attempts to piece together complete event information from fragmented messages.

Check Email and Teams

1. Student opens Microsoft Outlook or Microsoft Teams.
2. Student sorts through mass email blasts and group notifications.
3. Student searches for specific announcements among unrelated messages.
4. Student flags or saves relevant updates to revisit later.

Check Physical Bulletin Boards

1. Student visits high-traffic areas such as hallways or the cafeteria.
2. Student reads printed posters and flyers posted on corkboards.
3. Student notes event details manually or photographs the poster.

4. Student determines whether posted information is still current or outdated.

Manual Information Retrieval and Verification

1. Student asks classmates or organization members for updates through word-of-mouth.
2. Student cross-references information gathered from multiple platforms.
3. Student determines whether the collected information is complete and accurate.
4. Students decide whether to attend or participate based on available information.

SWOT Analysis

Strengths (Internal)

- **AI-Assisted Information Retrieval:** Retrieves and organizes campus-related announcements, events, and updates from multiple communication platforms into a single personalized mobile interface, helping reduce the effects of fragmented communication channels.
- **AI-Powered Assistance:** Features a built-in chatbot capable of summarizing lengthy announcements and answering user queries, which enhances the speed and convenience of information access.
- **Personalized Scheduling Support:** Helps students organize campus-related activities alongside their personal schedules, making it easier to identify events that match their availability.
- **Student-Centered Mobile Experience:** Tailored specifically for educational environments, providing students with a more organized and focused platform for accessing campus-related information compared to general social media applications.

Weaknesses (Internal)

- **Frontend-Focused Development Scope:** The current development team primarily focuses on the frontend mobile inquiry experience, while backend infrastructure and server management remain dependent on separate project components.

- **Connectivity Dependency:** The system requires a stable internet connection to retrieve and process information from authorized communication platforms and does not currently support offline functionality.
- **Functional Limitations:** The application focuses mainly on information retrieval and organization and does not include features such as private peer-to-peer messaging, institutional record management, or advanced administrative controls.
- **Initial Testing Scope:** Currently limited to Asia Pacific College (APC) as a testing environment, which may require additional adjustments before implementation in other educational institutions.

Opportunities (External)

- **Institutional Digital Transition:** Aligns with the growing adoption of AI-assisted and student-centered digital technologies within educational institutions.
- **Scalability for Other Schools:** Designed with a flexible framework that allows the mobile inquiry system to be adapted and implemented across multiple academic institutions beyond APC.
- **Support for Global Standards:** Supports United Nations Sustainable Development Goal 4 (Quality Education) and Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by improving digital access to educational and institutional information through mobile and AI-assisted technologies.
- **Increasing Demand for Information Accessibility:** Students increasingly require more efficient ways to manage fragmented campus communication and reduce the need to monitor multiple independent platforms.

Threats (External)

- **Established Communication Platforms:** Existing tools such as institutional email systems, messaging applications, and social media platforms are already deeply integrated into student communication habits.

- **User Permission and Privacy Concerns:** Since the system relies on retrieving information from authorized user accounts and communication platforms, concerns regarding privacy, permissions, and data security may affect user adoption.
- **Dependency on External Communication Sources:** The quality and completeness of retrieved information depend on the consistency and availability of announcements from existing communication platforms.
- **Rapid Changes in Platform Policies and APIs:** Changes in third-party platform policies, API accessibility, or integration limitations may affect the system's ability to retrieve and organize campus-related information consistently.

Proposed Solution

Technical Background

Student Profile Set-up

Before using the application, students are required to create a profile by providing information such as their full name, student ID number, course, year level, and contact details. During the setup process, students may also authorize selected communication platforms that the AI assistant may access for personalized information retrieval and organization.

The permission-based setup enables the system to retrieve campus-related announcements, schedules, and updates from authorized accounts according to the student's preferences while maintaining user-specific access control and data consistency.

Personalized Announcement Feed

NotifiED provides students with a personalized announcement experience by retrieving and organizing campus-related information from authorized communication platforms such as institutional email accounts and organization social media pages. Rather than requiring organizations to adopt a separate posting platform, the system functions as an AI-assisted retrieval layer that gathers relevant information and presents it through a structured mobile interface.

Retrieved announcements are organized according to relevance and user preferences, allowing students to access campus-related information more efficiently while reducing information fragmentation and platform ambiguity.

Announcement Filtering System

The application includes a filtering mechanism that enables students to organize retrieved announcements according to specific categories and areas of interest. Users may filter content by announcement type, events, academic schools, organizations, and other institutional classifications.

This feature allows students to quickly locate relevant information without manually reviewing large volumes of content, improving accessibility and information discovery within the platform.

Organization Directory

The Organization Directory allows students to explore recognized student organizations and campus groups through an organized interface. Each organization profile contains information such as its description, objectives, activities, officer information, contact details, and membership requirements.

To encourage student participation, the directory provides access to official organization resources, including membership application links and recruitment materials when available. This enables students to discover organizations and conveniently proceed to their respective application channels without searching across multiple platforms.

Automated Push Notifications

The application includes an automated push notification system that alerts students whenever relevant information is retrieved from authorized communication platforms. Instead of requiring users to actively search for updates, important announcements, event opportunities, and organization-related information are automatically delivered through personalized notifications.

This feature transforms information retrieval into a passive notification-based experience, helping students remain informed while reducing the likelihood of overlooking important campus updates.

AI-Assisted Chatbot

The AI-Assisted Chatbot serves as the interactive assistant component of the system. Through a conversational interface, students may ask questions regarding announcements, events, schedules, and other campus-related information retrieved from authorized communication platforms.

The chatbot analyzes retrieved information and provides summarized responses, simplified explanations of lengthy announcements, important reminders, and quick access to relevant details such as dates, venues, and event requirements. This enables students to understand campus-related information more efficiently without manually reviewing large amounts of content.

Schedule Integration

This feature allows students to voluntarily authorize access to supported academic scheduling platforms through a secure permission-based process. Upon user consent, the system retrieves schedule information such as lectures, laboratory sessions, meetings, and other academic commitments and displays them within the application.

By integrating existing academic schedules into the platform, students can view campus-related announcements and event opportunities alongside their academic commitments. This allows users to make more informed participation decisions while consolidating academic and campus-related information into a single personalized interface.

QR Code Event Registration

The application includes a QR code-based event registration feature that allows students to conveniently register for campus events through the mobile application. By scanning a QR code provided within event materials or authorized announcements, the system records the student's registration information using their stored profile details.

This feature simplifies the event participation process and serves as the final stage of the student engagement workflow, allowing students to transition directly from information discovery and event exploration to event registration within the same platform.

Feasibility

Operational Feasibility

- Students: NotifiED provides students with a centralized and personalized way of accessing campus-related information. Through AI-assisted retrieval,

announcement filtering, schedule integration, and automated notifications, students can stay informed without manually monitoring multiple communication platforms.

- **Permission-Based Information Retrieval:** The system operates through user-authorized access to supported communication and scheduling platforms. This allows students to control which accounts the application may access while enabling the retrieval and organization of relevant campus-related information within a single interface.
- **Compatibility with Existing Communication Channels:** Rather than replacing institutional communication platforms, the system works alongside existing channels by retrieving and organizing information from authorized sources. This minimizes disruption to current communication practices while improving accessibility for students.
- **Conclusion:** The system is operationally feasible because it enhances how students access, organize, and interact with campus-related information while remaining compatible with existing institutional communication ecosystems.

Economic Feasibility

- **Costs:** Development costs primarily involve mobile application development, cloud database hosting, AI integration services, and maintenance of platform integration components.
- **Reduced Information Search Time:** The system reduces the time students spend searching across multiple platforms for announcements, schedules, event details, and organization information.
- **Improved Information Accessibility:** AI-assisted retrieval, filtering, and summarization features improve the accessibility and organization of campus-related information.
- **Increased Student Engagement:** By simplifying access to events, organizations, and announcements, the platform may encourage greater participation in campus activities.

- Conclusion: The benefits of improved information accessibility, organization, and student engagement outweigh the costs associated with developing and maintaining the proposed mobile inquiry module.

Technical Feasibility

- Hardware: The application is designed for smartphones and mobile devices that are already commonly used by students, eliminating the need for additional specialized hardware.
- Software: The project utilizes established mobile development frameworks, cloud-based AI technologies, secure authentication mechanisms, and supported platform integration services. The successful implementation of schedule retrieval and account authorization demonstrates the feasibility of integrating existing communication and scheduling systems through permission-based access.
- Scalability: The architecture is designed to support integration with different institutional communication and scheduling platforms, allowing future adaptation across various educational institutions.
- Conclusion: The required technologies are mature, accessible, and supported by existing development frameworks, making the implementation of the proposed system technically feasible.

Schedule Feasibility

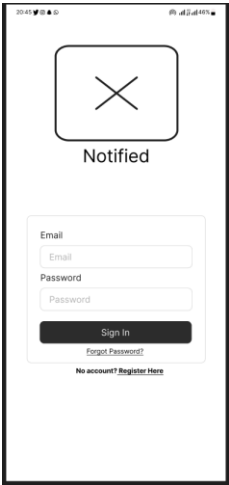
- Development Approach: The project follows a phased implementation strategy. Core features such as student profile setup, announcement retrieval, filtering mechanisms, and schedule integration are prioritized before advanced AI-assisted features and platform enhancements.
- Prototype Validation: Key integration components, including account authorization and schedule retrieval, will be demonstrated through a working prototype, reducing implementation uncertainty.
- Future Expansion: Additional integrations and AI capabilities can be incorporated incrementally without requiring major changes to the system architecture.

- Conclusion: The phased development approach and successful proof-of-concept implementations indicate that the proposed system can be completed within a standard academic project timeline while supporting future enhancements.

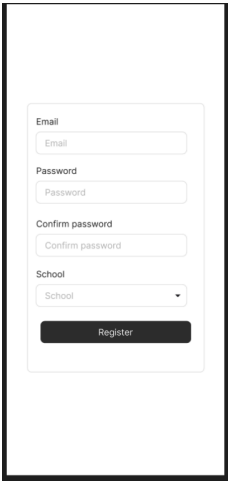
Requirements Analysis

Project Vision

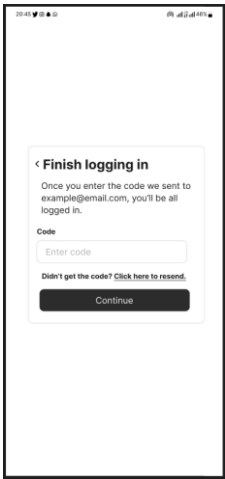
Prototype (Mock Flow / Wireframe)



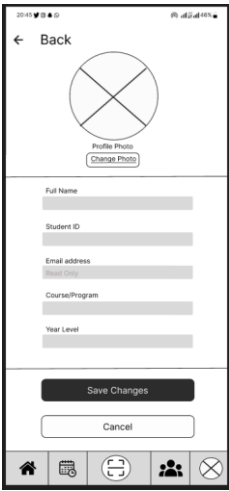
Login: Login Page



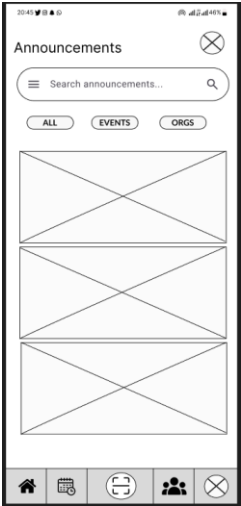
Registration: Students can register



OTP: Students receive OTP to login



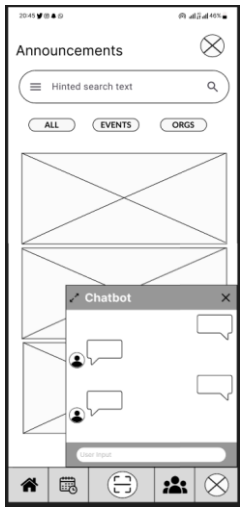
Student Profile: Students can update their information



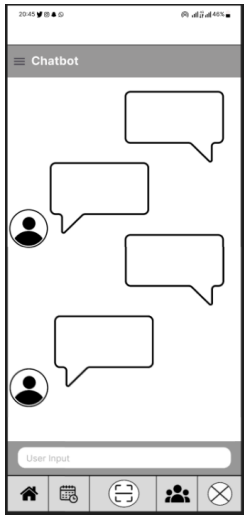
Announcement Page: All announcement and event page, students can also filter or search for easier navigation



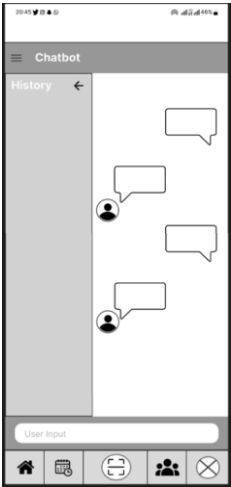
Detail Page: Students can see each announcement or event full details



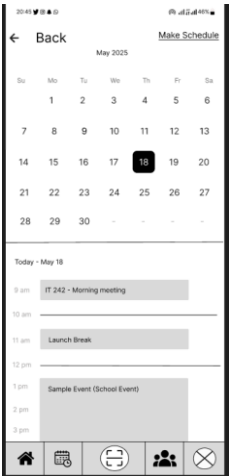
AI Chatbot (Floating): Floating AI chatbot



AI Chatbot (Expanded): Expanded AI chatbot



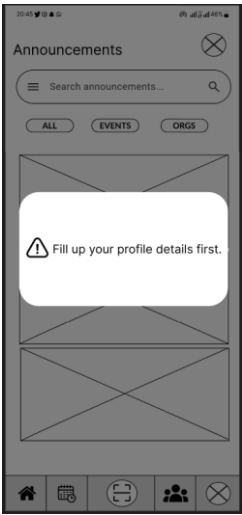
AI Chatbot (History): Chatbot History



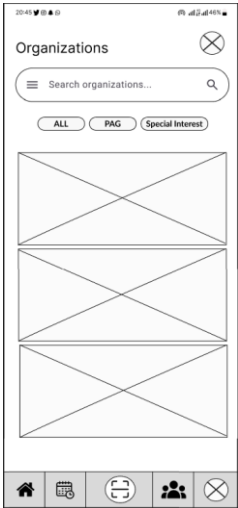
Schedule Management: Where students can see upcoming events and manage their own schedule



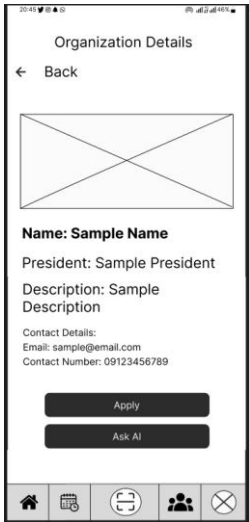
Schedule Management (Schedule Manager):
Where students manage their own schedule



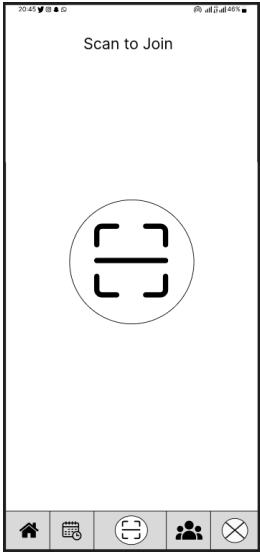
Warning: Warning pop-up when
students haven't filled up their profile details



Organization Directory: Students can view each organization and filter it



Organization Details: Details of each organization



QR Scanner: Students can register using QR scanner

Project Lean Canvas

Solution • AI-assisted retrieval of campus information from authorized communication platforms • Personalized mobile feed that organizes announcements, events, and organization updates • Intelligent student assistant with schedule integration, notifications, and event registration	Customer segments • Students • Educational Institutions Channels • Mobile application • University partnerships • Student ambassadors • Campus events and orientations • Social media promotion	Unique Value Proposition • AI-Assisted Personal Campus Information Hub	Key Metrics • Daily active users • Event registrations completed through the app • AI chatbot usage rate • Reduction in missed announcement reports Unfair Advantage • Permission-based information retrieval • Personalized information aggregation • AI-assisted information summarization • No platform migration required	Problem • Platform ambiguity • No unified access point • Event location difficulty Existing Alternatives • TV/monitors • Email platforms • Bulletin boards • Social media pages
Cost Structure • Mobile application development • Cloud hosting and database services • AI processing and API costs • Platform integration maintenance • Security and privacy compliance • Application maintenance and updates		Revenue Streams • Institutional licensing/subscription • Custom integration services for universities • Annual maintenance and support contracts		

User Classes and Characteristics

Roles	Description
Student	The primary end-user of the platform. This includes regular students who view announcements, browse events, and manage their schedules.
Notified campus system(web)	The centralized administrative interface. This component acts as the backend portal used by administrators and faculty advisers to review, approve, or disapprove event submissions, manage notifications, and monitor system analytics.

AI	The automated processing layer embedded within the system. This component generates summaries for announcements and events, answers student questions through the chatbot interface, and assists in detecting schedule conflicts.
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Product Backlog

User Story	Accepted Criteria
As a student, I want to view announcements in a chronological feed, so that I am always up to date with the latest official school news.	• Announcements are displayed in descending order, with the most recent entries shown first.
	• Each announcement includes a title, date, and a brief content preview.
	• A “Read More” option is available to display the full announcement details.
As a student, I want to ask the AI chatbot questions in natural language, so that I can get immediate assistance with campus-related inquiries	• The chatbot interface is accessible from the main application interface.
	• The system processes user input and generates contextually relevant responses.
	• The chatbot supports session continuity through chat history or provides an option to start a new session.
	• The system displays an error message if the user attempts to send a message without an active internet connection
	• Users can add, edit, and delete schedule entries with specific days and time slots.

As a student, I want to manage my class schedule by manually inputting my subjects, so that I have a digital copy of my daily timetable within the app	• The system stores schedule data and identifies potential conflicts with campus events.
	• The schedule is presented in a clear and structured daily or weekly format.
	• The system prevents the user from saving an entry if mandatory fields like "Subject Name" or "Time Slot" are left blank.
	• The system provides an "Undo" or "Confirm Delete" prompt before permanently removing a scheduled item to prevent accidental data loss.
As a student, I want to browse the organization directory so that I can discover different campus groups and find their contact information.	• A complete list of recognized campus organizations is displayed.
	• Selecting an organization reveals its details, including officers and contact information.
	• Events associated with each organization are accessible within the organization profile.

Product Roadmap

(Create your product roadmap. Template can be found in the CoursePack)

Release Plan

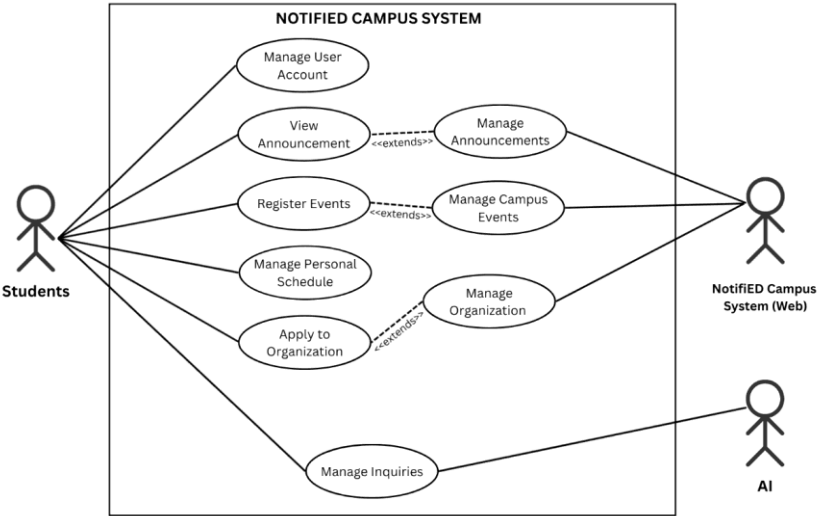
(

Include here your updated release plan and discuss whether you are on/ahead/behind schedule. Explain also how much of the product backlog has been finished (it is up to you how you will present it, without being repetitive. You

can say something like, “The complete product backlog is in Table ____”. Note: template can be found in the CoursePack.

)

Use Case Diagram



Use Case Full Description

Use Case 1: Manage User Account

Field	Description
Use Case Name	Manage User Account
Actor	Students

Description	The student creates and manages their personal profile within the NotifiED application, providing necessary information such as name, student ID, course, and year level.
Precondition	The student has downloaded and opened the NotifiED application.
Main Flow	1. Student opens the application. 2. Student is prompted to create or update their account. 3. Student enters required profile information. 4. System saves and confirms the account details.
Postcondition	The student's account is created and they are granted access to the application.

Use Case 2: View Announcement

Field	Description
Use Case Name	View Announcement
Actor	Students
Description	The student optionally opens and reads a specific announcement from the centralized announcement feed. This use case extends Manage Announcements.
Precondition	The student is logged in and the announcement feed is loaded.
Main Flow	1. Student browses the announcement feed. 2. Student selects a specific announcement. 3. System displays the full details of the selected announcement.
Postcondition	The student has read the full announcement details.

Extension Point	Extends Manage Announcements: triggered only when the student chooses to open a specific announcement.
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Use Case 3: Manage Announcements

Field	Description
Use Case Name	Manage Announcements
Actor	Students, NotifiED Campus System (Web)
Description	The system retrieves and displays all campus announcements from the web backend, providing students with a centralized and organized feed of official updates.
Precondition	The student is logged in and has a stable internet connection.
Main Flow	1. Student navigates to the announcements section. 2. System sends a request to the NotifiED Campus System (Web). 3. Web system returns the list of available announcements. 4. System displays the announcements in the student's feed.
Postcondition	The student can view all available campus announcements in one place.

Use Case 4: Register Events

Field	Description
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Use Case Name	Register Events
Actor	Students
Description	The student optionally registers for a specific campus event by scanning a QR code or manually inputting their information. This use case extends Manage Campus Events.
Precondition	The student is logged in and has accessed the campus events section.
Main Flow	1. Student browses available campus events. 2. Student selects an event they wish to join. 3. Student scans the QR code or input their information associated with the event. 4. System records the student's registration. 5. System confirms successful registration.
Postcondition	The student is successfully registered for the selected event.
Extension Point	Extends Manage Campus Events — triggered only when the student chooses to register for a specific event.

Use Case 5: Manage Campus Events

Field	Description
Use Case Name	Manage Campus Events
Actor	Students, NotifiED Campus System (Web)
Description	The system retrieves and displays all available campus events from the web backend, allowing students to explore upcoming activities and opportunities.

Precondition	The student is logged in and has a stable internet connection.
Main Flow	1. Student navigates to the campus events section. 2. System sends a request to the NotifiED Campus System (Web). 3. Web system returns the list of available events. 4. System displays the events to the student.
Postcondition	The student can view all available campus events.

Use Case 6: Manage Personal Schedule

Field	Description
Use Case Name	Manage Personal Schedule
Actor	Students
Description	The student manually inputs and manages their personal class schedule within the application, which the AI then uses to identify free time slots and suggest productive windows.
Precondition	The student is logged in and has accessed the schedule section.
Main Flow	1. Student navigates to the personal schedule section. 2. Student manually inputs their class schedules, including subject, time, and day. 3. System saves and displays the student's weekly schedule. 4. AI analyzes the schedule and identifies available time slots.
Postcondition	The student's personal schedule is saved and visible within the application.

Use Case 7: Apply to Organization

Field	Description
Use Case Name	Apply to Organization
Actor	Students
Description	The student optionally submits an application to join a campus organization after exploring the organization directory. This use case extends Manage Organization.
Precondition	The student is logged in and has accessed the organization directory.
Main Flow	1. Student browses the organization directory. 2. Student selects an organization they wish to join. 3. Student submits an application through the platform. 4. System records and forwards the application. 5. System confirms that the application has been submitted.
Postcondition	The student's application has been successfully submitted to the selected organization.
Extension Point	Extends Manage Organization — triggered only when the student chooses to apply to a specific organization.

Use Case 8: Manage Organization

Field	Description
Use Case Name	Manage Organization
Actor	Students, NotifiED Campus System (Web)

Description	The system retrieves and displays the organization directory from the web backend, providing students with a centralized listing of all recognized campus organizations and their details.
Precondition	The student is logged in and has a stable internet connection.
Main Flow	1. Student navigates to the organization directory. 2. System sends a request to the Notified Campus System (Web). 3. Web system returns the list of organizations and their profile details. 4. System displays the organizations to the student.
Postcondition	The student can explore all available campus organizations and their information.

Use Case 9: Manage Inquiries

Field	Description
Use Case Name	Manage Inquiries
Actor	Students, AI
Description	The student submits a question or inquiry through the conversational AI interface, and the AI processes the request by scanning the platform database to provide an immediate and accurate response or summary.
Precondition	The student is logged in and has accessed the inquiry or chatbot section.
Main Flow	1. Student opens the AI chatbot interface. 2. Student types and submits a question or inquiry. 3. AI scans the platform database for relevant information. 4. AI returns a response, summary, or actionable answer to the student. 5. System displays the AI's response in the chat interface.

Postcondition	The student has received a relevant response to their inquiry from the AI.
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Conclusion

This project established the foundation for addressing the challenges of fragmented campus communication and platform ambiguity within educational institutions, using Asia Pacific College (APC) as the primary context. Through research, system planning, requirements analysis, and system design, the NotifiED framework was developed as an AI-assisted mobile platform intended to consolidate campus-related information from multiple communication channels into a centralized and personalized student experience.

During this initial design phase, the project successfully achieved the following objectives:

- **Design of a Centralized Information Retrieval Framework:** To address the issue of platform ambiguity, a system architecture was developed that retrieves and organizes campus-related information from authorized communication platforms. This framework provides students with a unified access point for announcements, events, and organizational updates without requiring institutions or organizations to migrate to a separate posting platform.
- **Development of Personalized Information Access Mechanisms:** To improve information accessibility and reduce information fragmentation, the project designed a personalized announcement feed, announcement filtering system, automated notification mechanism, and organization directory. These features enable students to efficiently discover and access relevant campus information according to their interests and academic needs.
- **Integration of AI-Assisted Student Support Features:** To enhance information comprehension and engagement, the proposed solution incorporates an AI-assisted chatbot capable of helping students interact with retrieved campus information through natural language inquiries. This allows users to obtain summarized responses and quickly access important details regarding announcements, events, schedules, and organizational activities.

- **Design of Student Engagement and Participation Features:** To support student involvement beyond information discovery, the system incorporates organization exploration, schedule integration, and QR code-based event registration features. These components create a more complete student engagement workflow by connecting information access with participation opportunities.

Overall, the project demonstrates the potential of an AI-assisted campus information hub to improve how students discover, organize, and interact with campus-related information. While further development, testing, and institutional collaboration are necessary for full implementation, the proposed framework provides a strong foundation for addressing communication inefficiencies and improving student access to relevant campus information.

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Appendices

Appendix A: Roles and Responsibilities

Project leader: Charles Capagngan

Responsibilities: Led overall project coordination and documentation. Contributed to wireframes and prototype mock flows. Participated in Design Thinking (UCD and Lean Canvas).

Member: Cristian Saldua

Responsibilities: Assisted in documentation and contributed to wireframes and prototype mock flows. Participated in Design Thinking (UCD and Lean Canvas).

Member: Gabrielle Calibo

Responsibilities: Contributed to project documentation. Participated in Design Thinking (UCD and Lean Canvas).

Member: Patricia Marie Valera

Responsibilities: Contributed to project documentation. Participated in Design Thinking (UCD and Lean Canvas).

Appendix B: Minutes of the Meetings

Link: [Minutes of the Meetings](#)

Meeting with Sir Gonzalo Gumogda

Minutes of Meeting

Project Name: NotifiED: AI-Assisted Campus Engagement Framework

Date: 25/03/2026

Time: 4:00 pm – 5:00 pm

Location/Platform: Microsoft Teams Meeting

Attendees

- **Present:** Charles Capagngan, Cristian Saldua, Gabrielle Calibo, Patricia Marie Valera

Agenda Items

A. Project Overview

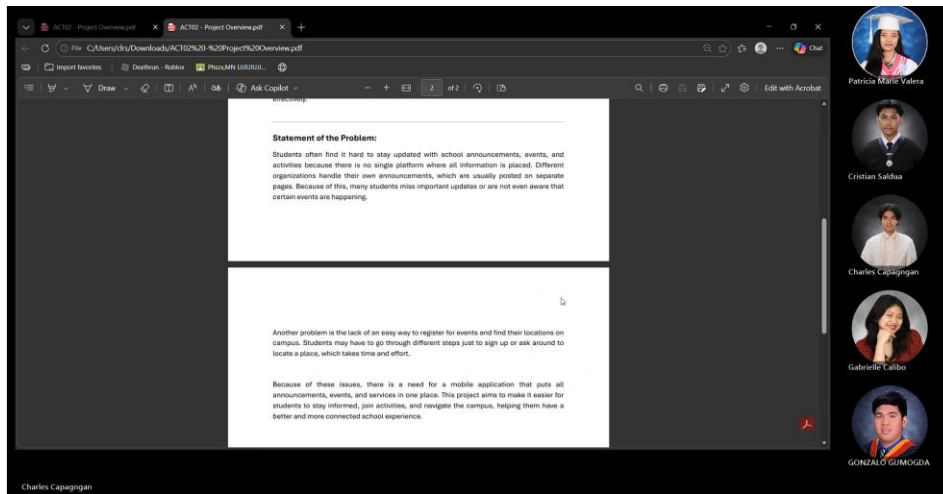
- **Discussion:** The Mobile Group presented the overall vision of the proposed project (*NotifiED*) to their adviser. The adviser emphasized that the Statement of the Problem (SOP) should not be generalized. Instead, it must clearly reflect the specific objectives, scope, and unique contributions of the mobile group within the *NotifiED* project. The adviser highlighted the importance of tailoring the SOP to the actual needs of the campus community and ensuring it demonstrates the distinct value of mobile implementation.
- **Outcome:** The group acknowledged the adviser's feedback and agreed to revise the SOP to make it more specific, detailed, and aligned with the project's goals.

Action Items (To-Do List)

- Revise the SOP to avoid generalized statements.

Other Business

- None.



Commented [GC1]: 25/03 w sir gonz

Meeting with Ms Ma Revvie Zoe Congson

Minutes of Meeting

Project Name: NotifiED: AI-Assisted Campus Engagement Framework

Date: 31/03/2026

Time: 1:30 pm – 2:30 pm

Location/Platform: Microsoft Teams Meeting

Attendees

- **Present:** Charles Capagngan, Cristian Saldia, Gabrielle Calibo, Patricia Marie Valera

Agenda Items

A. Introduction of the Project

- **Discussion:** The group introduced the project overview, highlighting the communication issues faced by students. The adviser confirmed that the issue is valid:
 - Some organizations only publish selected events, leading to incomplete visibility.
 - Certain event details overlap, causing confusion.
 - Multiple platforms are used for announcements, and with the large number of events, some information gets buried and overlooked.
- **Outcome:** Ms. Revvie acknowledged the relevance of the problem

B. Collaboration with Other Groups

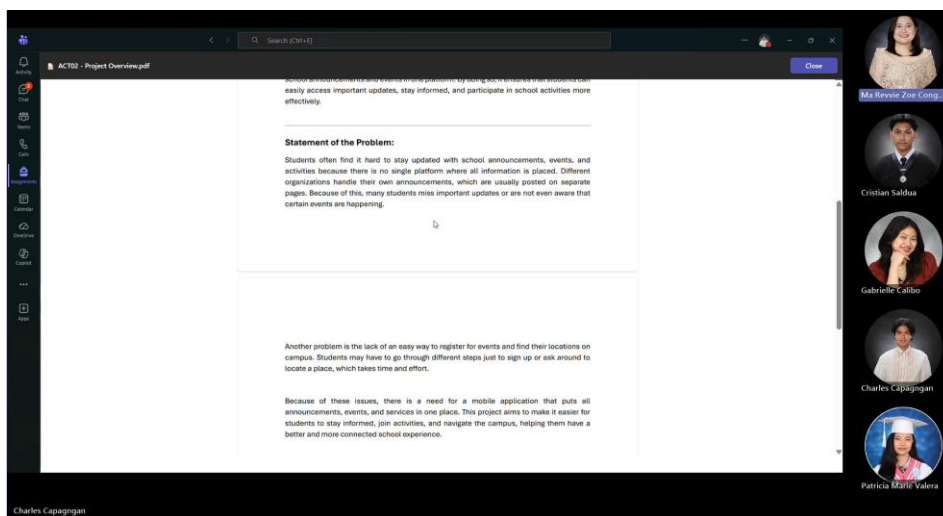
- **Discussion:** The group decided to create a group chat with another team under Sir Dom to strengthen collaboration and ensure alignment across related projects.
- **Outcome:** The group set up the group chat and coordinated with Sir Dom's team.

Action Items (To-Do List)

- Create a group chat with Sir Dom's team for collaboration.
- Update project documentation to Ms. Revvie's feedback.

Other Business

- None



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Meeting with Sir Gonzalo Gumogda

Minutes of Meeting

Project Name: NotifiED: AI-Assisted Campus Engagement Framework

Date: 01/04/2026

Time: 2:30 pm – 2:40 pm

Location/Platform: Microsoft Teams Meeting

Attendees

- **Present:** Charles Capagnan
- **Absent:** Cristian Saldua, Gabrielle Calibo, Patricia Marie Valera

Agenda Items

A. Update on Meeting with Ms. Revvie Zoe Congson

- **Discussion:** The Mobile Group updated the research adviser regarding the outcomes of the meeting with Ms. Revvie. Key points shared included:
 - Validation of the project's identified issues (incomplete visibility of events, overlapping details, and announcement overload across multiple platforms).
 - Agreement to emphasize these issues in the project documentation.
 - Decision to establish a group chat with another team under Sir Dom to strengthen collaboration.
- **Outcome:** The research adviser acknowledged the updates and advised the group to integrate Ms. Revvie's feedback into the ongoing project documentation and coordination efforts.

Other Business

- None.

Meeting with Sir Gonzalo Gumogda

Minutes of Meeting

Project Name: Notified: AI-Assisted Campus Engagement Framework

Date: 08/04/2026

Time: 2:30 pm – 3:30 pm

Location/Platform: Microsoft Teams Meeting

Attendees

- **Present:** Charles Capagngan, Cristian Saldua, Gabrielle Calibo
- **Absent:** Patricia Marie Valera

Agenda Items**A. Class Schedule Feature**

- **Discussion:** The adviser noted that since class schedules are already available in Microsoft Teams, the feature within the app may be redundant. However, it was acknowledged as acceptable if included for completeness.
- **Outcome:** The group agreed to deprioritize the class schedule feature but keep it documented as optional.

B. Lean Canvas Clarification

- **Discussion:** The adviser reviewed Lean Canvas and suggested fixing the distinction between school and student organizations. Specifically, school events should be clearly attributed to the school administration, not the organizations.
- **Outcome:** The group agreed to update the Customer Segment to reflect this distinction and ensure consistency across the canvas.

C. Survey Questions Draft

- **Discussion:** The group drafted initial survey questions for students to validate assumptions about event participation and communication challenges.
- **Outcome:** The adviser approved the direction but requested refinement of wording for clarity.

Action Items (To-Do List)

- Update Lean Canvas
- Refine survey questions for clarity and adviser approval.
- Document optional inclusion of class schedule feature.

Other Business

- None

Meeting with Web Group

Minutes of Meeting

Project Name: NotifiED: AI-Assisted Campus Engagement Framework

Date: 15/04/2026

Time: 1:00 pm – 2:00 pm

Location/Platform: Microsoft Teams Meeting

Attendees

- **Present:** Charles Capagngan, Cristian Saldua, Gabrielle Calibo, Patricia Marie Valera [Web Group Representatives]

Agenda Items

A. Joint Group Integration and Gap Analysis

- **Discussion:** The Mobile Group met with the Web Group to share updates and identify gaps in development. From the mobile perspective, synchronization issues were noted, particularly in aligning features and ensuring consistent user experience across platforms.
- **Outcome:** Agreed on a list of improvements to strengthen cohesion and ensure seamless integration between mobile and web versions.

B. Scope Expansion: Student Life Organization

- **Discussion:** The Mobile Group supported the proposal to include features for organizing student life activities, noting its importance for mobile users who rely on quick access to campus-related events.
- **Outcome:** Student Life Organization formally added to project objectives, with mobile group tasked to draft requirements for mobile-specific use cases.

C. Project-Based Learning (PBL) Synchronization

- **Discussion:** The Mobile Group shared its PBL goals and clarified how these contribute to the centralized directory system. Alignment with the Web Group was discussed to maintain a unified vision under adviser guidance.
- **Outcome:** Both groups agreed to synchronize PBL requirements to ensure consistency.

D. AI Implementation: Event Summarization

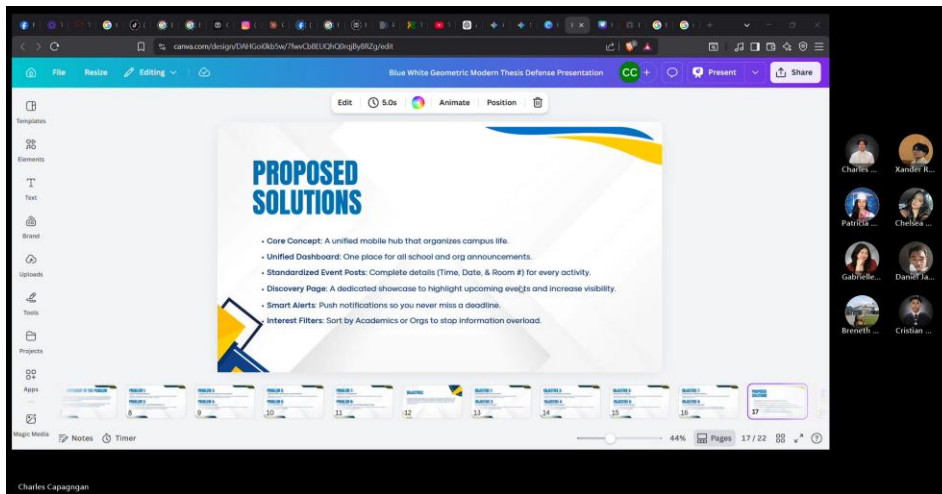
- **Discussion:** The Mobile Group expressed interest in integrating AI assistants to help students quickly digest event details. This was seen as a key feature for mobile usability.
- **Outcome:** AI summarization formally included.

Action Items (To-Do List)

- Draft technical requirements for the Student Life Organization module (mobile focus).
- Research and select AI models for mobile event summarization.
- Document PBL contributions from the Mobile Group.
- Coordinate integration points with the Web Group for synchronization.

Other Business

- None



Meeting with Sir Gonzalo Gumogda

Minutes of Meeting

Project Name: NotifiED: AI-Assisted Campus Engagement Framework

Date: 16/04/2026

Time: After class at 5:30 pm

Location/Platform: Face to Face

Attendees

- **Present:** Charles Capagngan, Cristian Saldua
- **Absent:** Gabrielle Calibo, Patricia Marie Valera

Agenda Items

A. Customer Segment Update

- **Discussion:** The adviser recommended refining the customer segment to focus on Senior High School, Colleges, and Universities, and removing “event organizer” as a separate category.
- **Outcome:** The group agreed to update the Lean Canvas accordingly.

B. Problem Consolidation

- **Discussion:** Statement of the Problem 2–6 and 8 were reviewed and determined to overlap. Adviser suggested merging them into one consolidated problem statement for clarity.
- **Outcome:** The group will reframe these into a single, unified problem in the Lean Canvas.

C. Umbrella Question

- **Discussion:** Adviser emphasized the need to include an umbrella question in the survey to capture overarching student challenges.
- **Outcome:** The group will draft and integrate a broad, guiding question into the survey.

D. Objective Refinement

- **Discussion:** Objective 8 was identified as needing rephrasing to improve clarity and alignment with project goals.
- **Outcome:** The group will revise Objective 8 into a more precise statement.

E. Project Limitations

- **Discussion:** Adviser confirmed that the project will be limited to a web application implementation.
- **Outcome:** Limitation noted in the project scope documentation.

F. Feature Addition

- **Discussion:** A new feature was proposed where students signing in will bind their accounts to social media for easier integration.
- **Outcome:** Feature added to the technical requirements list.

G. AI Integration

- **Discussion:** Adviser suggested incorporating AI work scheduling and an AI chatbot into the project scope to enhance usability.
- **Outcome:** AI features formally included in the project objectives.

Action Items (To-Do List)

- Update Customer Segment (remove event organizer, focus on SHS, Colleges, Universities).
- Consolidate overlapping problems into one unified statement.
- Draft and include an umbrella survey question.
- Rephrase Objective 8 for clarity.
- Document limitation: web application
- Add social media binding features to requirements.
- Research and plan AI work scheduling and chatbot integration.

Other Business

- None

Meeting with Web Group

Minutes of Meeting

Project Name: NotifiED: AI-Assisted Campus Engagement Framework

Date: 21/04/2026

Time: 8:00 pm – 8:30 pm

Location/Platform: Microsoft Teams Meeting

Attendees

- **Present:** Charles Capangnan, Cristian Saldua, Gabrielle Calibo, Patricia Marie Valera, and [Web Representative]

Agenda Items

A. Paper Revisions

- **Discussion:** Both groups shared the revisions made to their project papers. The Mobile Group highlighted updates in the statement of the problem, and refined objectives based on adviser's feedback. The Web Group presented adjustments in Demographics and Information Fragmentation making it aligned with Mobile Group.
- **Outcome:** Both groups agreed to ensure consistency across documents and align revisions for a unified project presentation.

B. AI Implementation and Features

- **Discussion:** The Mobile Group discussed potential AI features to be implemented in the mobile app. Proposed features included:
 - **AI Chatbot** – designed to summarize posted events within the app. Its role is to highlight which events students can attend and provide the essential event information in a clear, accessible format.
 - **AI Class Scheduling** – envisioned as a smart scheduler. When an event is posted, the AI can automatically assign it to a student's schedule. If the event conflicts with an existing class schedule, the AI will not assign it, ensuring no overlaps.
 - **AI Event Summarization** – transforming event posts into short summaries with essential information.
- **Outcome:** The research adviser acknowledged the updates and advised the group to integrate Ms. Revvie's feedback into the ongoing project documentation and coordination efforts.

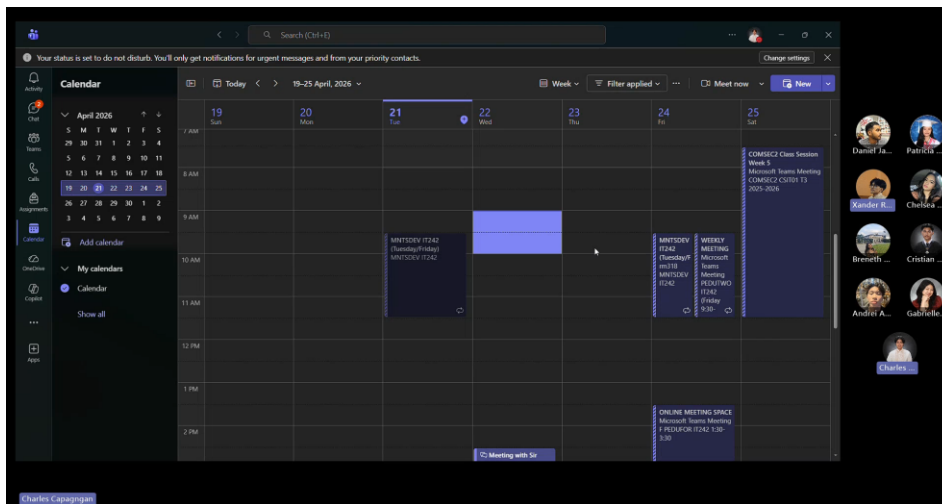
C. Mobile Group Update: App Life Cycle and Features

- **Discussion:** The Mobile Group updated the Web Group that they had already brainstormed the life cycle of the app, outlining the following features and flow:
 - **User Login** – Secure entry point for students and faculty.
 - **Landing Page** – Primary dashboard after authentication.
 - **Announcement Feed** – Central hub for school news and urgent updates.
 - **AI Chatbot** – Floating support icon for instant inquiries.

- **Bottom Navigation** – Persistent menu to toggle between modules:
 - **Class Schedule** – Manual input and tracking of daily subjects.
 - **User Profile** – Manage account settings and create feedback.
 - **School Organizations & Activities** – Org Directory: A list of all school organizations including Contact Persons.
- **Event Registration** – Register for and view upcoming activities.
- **Outcome:** Web Group acknowledged the Mobile Group's progress and agreed to align backend dependencies with the proposed mobile features.

Other Business

- None.



Meeting with Sir Gonzalo Gumogda

Minutes of Meeting

Project Name: NotifiED: AI-Assisted Campus Engagement Framework

Date: 22/04/2026

Time: 2:30 pm to 3:30 pm

Location/Platform: Microsoft Teams Meeting

Attendees

- **Present:** Charles Capagngan, Cristian Saldua, Gabrielle Calibo
- **Absent:** Patricia Marie Valera

Agenda Items

A. AI Implementation Updates

- **Discussion:** The Mobile Group updated the research adviser on the AI features being considered for implementation. These included event summarization, chatbot integration, and class scheduling.
- **Outcome:** Adviser acknowledged the progress and encouraged the group to refine the scope of AI features to ensure practical usability.

B. B. Chatbot Functionality

- **Discussion:** The research adviser suggested enhancing the chatbot so that students could register for events directly through it. This would allow the chatbot not only to summarize events but also to facilitate participation.
- **Outcome:** The group agreed to explore this functionality as a potential enhancement, recognizing its value for improving student engagement and simplifying event registration.

C. Revision of Objective 3

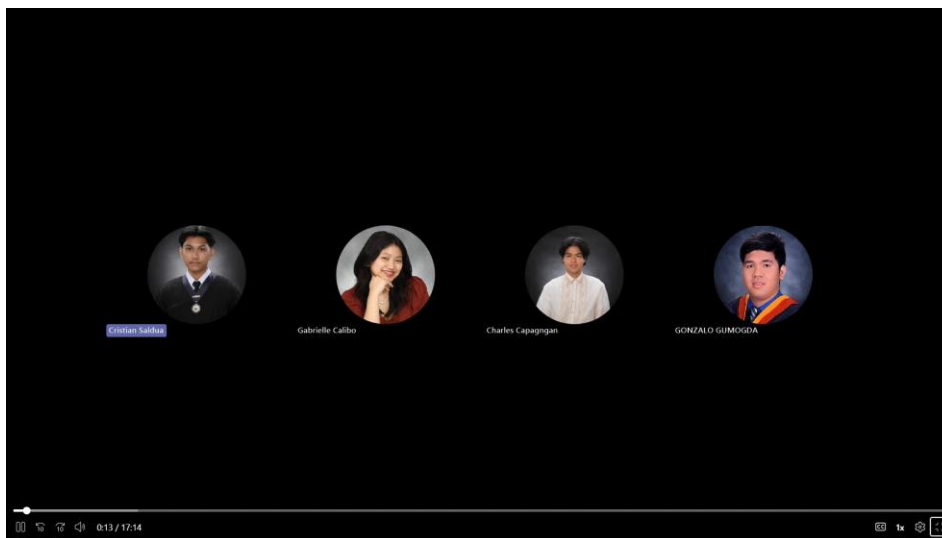
- **Discussion:** The adviser recommended revising Objective 3 in the project paper to better align with the statement of the problem.
- **Outcome:** The Mobile Group committed to rephrasing Objective 3 and ensuring consistency with the overall project objectives.

Action Items

- Explore feasibility of chatbot-based event registration.
- Revise Objective 3 in the project paper to reflect adviser's feedback.

Other Business

- None.



Meeting with Web Group

Minutes of Meeting

Project Name: NotifiED: AI-Assisted Campus Engagement Framework

Date: 08/05/2026

Time: 8:30 pm to 9:00 pm

Location/Platform: Microsoft Teams Meeting

Attendees

- **Present:** Charles Capagngan, Cristian Saldua, Gabrielle Calibo, Patricia Marie Valera, Web Representatives

Agenda Items

A. Use Case Diagram

- **Discussion:** Both groups collaborated on the creation of the use case diagram. The Mobile Group raised points on how their use cases would connect with the Web Group's system flow.
- **Outcome:** Agreed to align mobile and web use cases to ensure seamless integration.

B. User Story

- **Discussion:** The groups discussed user stories to capture how students and faculty would interact with the app.
- **Outcome:** Draft user stories were created, with refinement to follow based on adviser feedback.

C. Product Teaser Video

- **Discussion:** The teaser video was reviewed, covering all features for both mobile and web groups.
- **Outcome:** Both groups confirmed that the teaser video successfully showcased the app's core features.

D. Return on Investment (ROI)

- **Discussion:** ROI was discussed, noting that for now, there are no expenses incurred by either group.
- **Outcome:** ROI documentation will reflect zero current costs, with future updates to be added once expenditures arise.

E. Templates for AI Chatbots

- **Discussion:** The groups explored approaches for creating the AI chatbot. Current research is focused on available frameworks, with OpenAI identified as the initial platform to explore.
- **Outcome:** Agreed to continue researching chatbot templates and test OpenAI for feasibility.

F. Templates for AI Chatbots

- **Discussion:** The Mobile Group raised the possibility that if Sir Dom does not approve assigning the admin role solely to the Web Group, both groups will share the responsibility equally (50/50).
- **Outcome:** Both groups agreed to prepare for a shared admin role setup if required.

Action Items

- Align mobile use cases with Web Group's diagram for integration.
- Refine mobile-specific user stories.
- Ensure mobile features are properly represented in the teaser video.
- Document ROI status (no current expenses).
- Research chatbot templates and explore OpenAI for mobile implementation.
- Prepare for potential shared admin role (50/50) with Web Group.

Other Business

- None.

	A	B	C	D	E
1	Members:	Xander Revelar, Breneth Jan Ananayo	Chelen Hillary Nacalaban, Daniel James Whitwell	Andrei Antonio Villa	
2	User Case:	Notified Events & Activity List			
3					
4	User Phase	User Story	Accepted Criteria		
5	Activities and Events Discovery	As a User, I want to view a list of all statuses and upcoming activities and events so that I can stay informed about scheduled activities and events.	• Display a chronologically ordered list of activities and events. • Show activities and events by date, time, place, and status. • Include a search bar to filter activities and events by name.		
6	Notification of Activities and Events	As a User, I want to receive notifications for new and upcoming activity and event updates so that I don't miss important changes.	• A notification badge appears when new activities and events are added or updates are made. • Users can click the notification to be directed to the activity and event details. • Push notifications are sent to the web application.		
		As a User, I want to filter activities by school program	• The system provides tag-based filtering for school programs (e.g., Information Technology, Computer Science, Engineering, Business, Multimedia Arts).		

Meeting with Sir Gonzalo Gumogda

Minutes of Meeting

Project Name: Notified: AI-Assisted Campus Engagement Framework

Date: 13/05/2026

Time: 2:30 pm to 3:00 pm

Location/Platform: Microsoft Teams Meeting

Attendees

- **Present:** Charles Capagngan, Cristian Saldua, Gabrielle Calibo, Patricia Marie Valera,

Agenda Items

A. Admin Role Concern

- **Discussion:** The research adviser clarified that the mobile group should not worry about the admin role.
- **Outcome:** Mobile group agreed to set aside admin role concerns and focus on core deliverables.

B. Use Case Diagram Alignment

- **Discussion:** Adviser emphasized that the UCD should be based directly on the user stories to ensure consistency and relevance.
- **Outcome:** The group agreed to refine the UCD by mapping it strictly to the user stories.

C. User Use Cases

- **Discussion:** Adviser and the group narrowed down that there should be four use cases for the user to keep the scope clear and manageable.
- **Outcome:** The group committed to finalizing four distinct user use cases in the UCD.

D. Integration with ITRO and MS 365

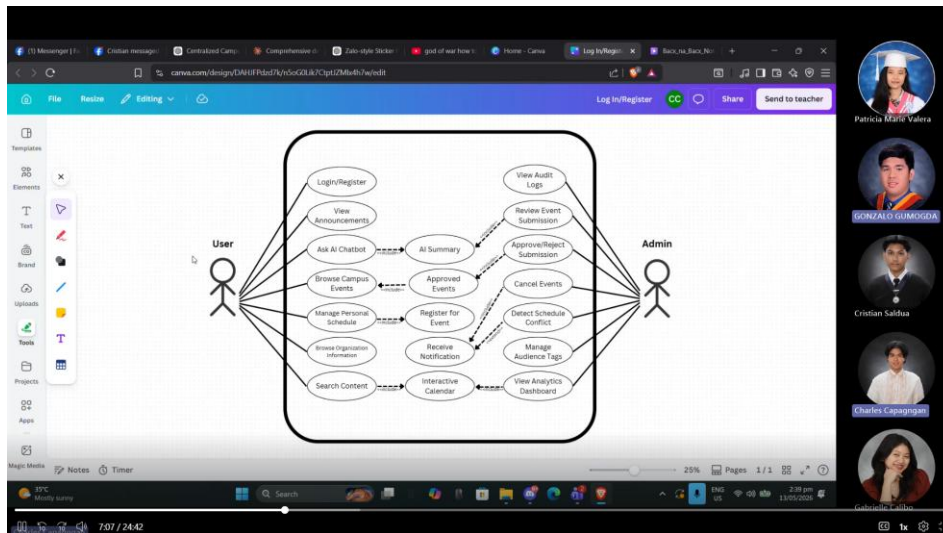
- **Discussion:** Adviser suggested reaching out to ITRO to explore the possibility of integrating the app with Microsoft 365 for access and authentication.
- **Outcome:** The group agreed to initiate communication with ITRO regarding MS 365 integration.

Action Items

- Refine the UCD by aligning it with user stories.
- Set aside admin role concerns per adviser's guidance.
- Coordinate with ITRO to explore MS 365 integration feasibility.

Other Business

- None.



Meeting with Web Group

Minutes of Meeting

Project Name: NotifiED: AI-Assisted Campus Engagement Framework

Date: 13/05/2026

Time: 8:00 pm to 8:30 pm

Location/Platform: Microsoft Teams Meeting

Attendees

- **Present:** Cristian Saldua, Patricia Marie Valera, Web Representatives
- **Absent:** Charles Capagngan, Gabrielle Calibo

Agenda Items

A. Feedback from Professor on Use Case Diagram and Web App

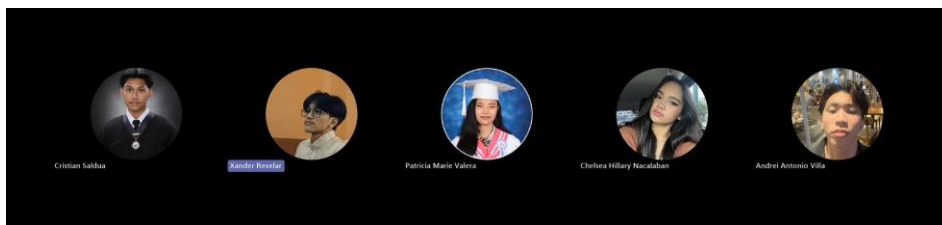
- **Discussion:** The Mobile Group informed the Web Group about the professor's comments regarding the Use Case Diagram (UCD) and the web app. According to the professor and the Mobile Group's research adviser, the current web app design was incorrect.
- **Outcome:** Both groups acknowledged the feedback

B. Use Case Diagram Alignment

- **Discussion:** The Web Group thought of expanding the scope by creating a web version as well, to cover a wider environment and ensure accessibility beyond mobile.
- **Outcome:** The Mobile Group clarified that the Web Group does not need to create a separate web app according to the professor, since their role is focused purely on backend development.

Other Business

- None.



Meeting with Web Group

Minutes of Meeting

Project Name: NotifiED: AI-Assisted Campus Engagement Framework

Date: 19/05/2026

Time: 9:40 am to 10:00 am

Location/Platform: Microsoft Teams Meeting

Attendees

- **Present:** Charles Capagnan, Cristian Saldua
- **Absent:** Gabrielle Calibo, Patricia Marie Valera

Agenda Items

A. Presentation Content

- **Discussion:** The Mobile Group discussed the content of the PowerPoint presentation.
- **Outcome:** The group agreed to revise the UCD accordingly.

B. Survey Support for Problem Statement and Survey Results

- **Discussion:** The Mobile Group emphasized the importance of the survey results to support the statement of the problem and reviewed the survey findings and analyzed how they reflect student and faculty needs.
- **Outcome:** Web Group acknowledged the survey's role in strengthening the research foundation. The results will be integrated into the presentation and documentation to validate the problem statement.

C. Lean Canvas Model

- **Discussion:** Focused discussion on the Lean Canvas, particularly the Cost Structure and Revenue Stream sections.
- **Outcome:** The discussion served as groundwork for future refinement.

D. UCD and User Story

- **Discussion:** The Mobile Group presented their Use Case Diagram (UCD) and user story to the Web Group.
- **Outcome:** Feedback was noted to ensure consistency across mobile and backend perspectives.

E. App UI Concept

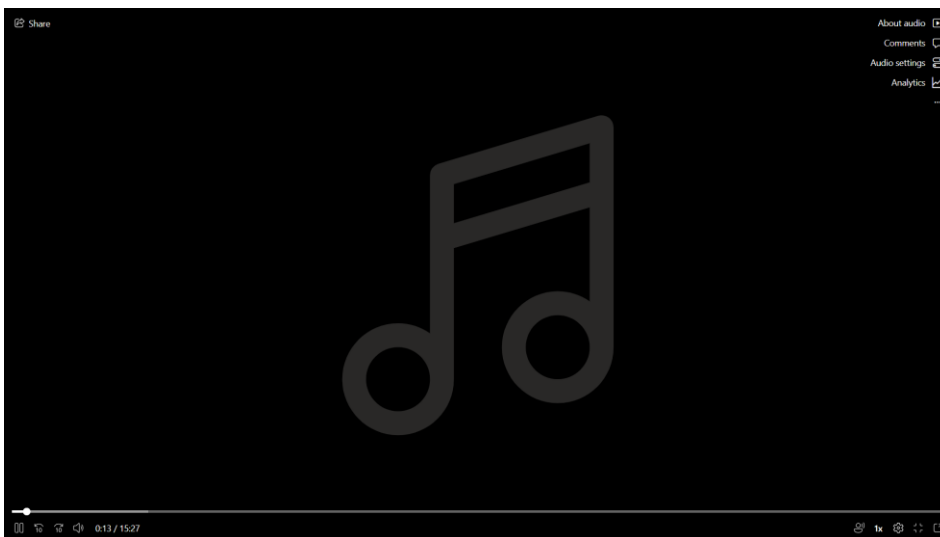
- **Discussion:** The Mobile Group showed and discussed their idea for the app's user interface, highlighting navigation flow and feature accessibility.
- **Outcome:** Web Group acknowledged the design.

Action Items

- Refine PowerPoint content further if needed.
- Integrate survey results into the problem statement and presentation.
- Document survey findings in project paper.
- Finalize UCD and user story for inclusion in deliverables.

Other Business

- None.



Meeting with Sir Gonzalo Gumogda

Minutes of Meeting

Project Name: NotifiED: AI-Assisted Campus Engagement Framework

Date: 20/05/2026

Time: 2:00 pm to 2:30 pm

Location/Platform: Microsoft Teams Meeting

Attendees

- **Present:** Cristian Saldua, Charles Capagnan, Gabrielle Calibo
- **Absent:** Patricia Marie Valera

Agenda Items

A. Use Case Diagram (UCD) Suggestions

- **Discussion:** The Mobile Group relayed the research adviser's suggestions about the UCD and proposed fixes.
- **Outcome:** The group agreed to revise the UCD accordingly.

B. Product Increment and Security Enhancement

- **Discussion:** The adviser recommended adding a password requirement, noting that student accounts would not be secure if someone else obtained their email.
- **Outcome:** The team agreed to integrate password functionality into the design.

C. User Story Updates

- **Discussion:** The Mobile Group presented the user story, showing how users could manage inquiries connected to the UCD. The user story "I want to search content" was reworded to clarify cross-platform functionality.
- **Outcome:** User stories were updated and aligned with the UCD.

D. Web Group UCD Review

- **Discussion:** The Mobile Group reviewed the UCD created by the Web Group.
- **Outcome:** Feedback was provided to align both diagrams with adviser's guidance.

E. Reviewed and Accepted Deliverables

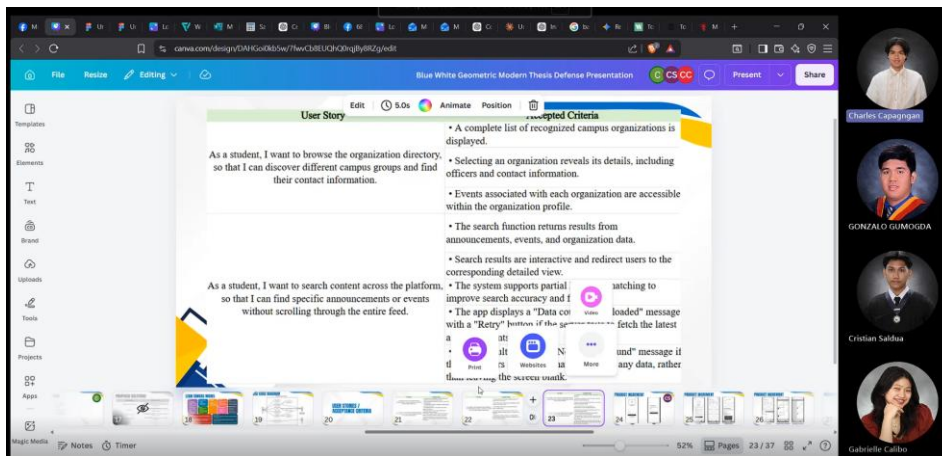
- **Discussion:** Both groups discussed the Lean Canvas model, proposed solutions, survey design, and the draft of the full paper description. The teaser video was also presented.
- **Outcome:** The research adviser confirmed that all these items were good as presented, requiring no further revisions at this stage.

Action Items

- Revise UCD based on adviser's suggestions and remove Admin/Faculty role.
- Update user stories and ensure alignment with UCD.
- Integrate password functionality into the design for security.
- Provide feedback on Web Group's UCD for consistency.
- Maintain Lean Canvas, proposed solutions, survey, paper draft, and teaser video as approved by adviser.

Other Business

- None.



Appendix C: Methodology

Survey

A quantitative survey design was utilized to investigate student experiences, challenges, and preferences regarding campus communication. The primary objective of this survey was to assess the impact of decentralized information across multiple platforms and to gauge student interest in potential solutions that NotifiedED can provide.

The primary data collection tool was a structured questionnaire deployed electronically. The section evaluated in this document consists of 10 distinct Likert-scale statements designed to measure student sentiment across three main areas:

- Information Fragmentation: Challenges with keeping track of announcements spread across multiple platforms (e.g., Facebook, Messenger, Email).
- User Experience & Engagement: Feelings of overwhelm due to notification fatigue and difficulty locating specific event details or campus venues.
- Solution Readiness: Assessment of student willingness to adopt a dedicated mobile app or an AI-powered summary tool.

Measurement Scale

Participants rated their level of agreement with each statement using a 5-point Likert Scale, ordered as follows:

1. Strongly Agree
2. Agree
3. Neutral
4. Disagree
5. Strongly Disagree

Survey Link:

https://forms.office.com/Pages/DesignPageV2.aspx?origin=NeoPortalPage&subpage=design&id=5kW3rmaBj0-SMxeegQnEnnrQ8nAqodAtaEP_349sUdUNU5ZVUdaRzczRTIFUVIMV1czQTJUV1pZWC4u&analysis=true

Appendix D: Project Sharepoint Link

<https://asiapacificcollege.sharepoint.com/:u:/r/sites/RamTrack-CharlesAngel%27s/SitePages/ProjectHome.aspx?csf=1&web=1&e=BNWR0U>

Appendix E: Requirements Traceability Matrix**Appendix F: RACI Matrix**